Uniform Bernstein–Sato Factors and Contact Roots for Surface Germs with a Multiple Smooth Tangent Cone

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20 October 2026

Abstract

Let $f = g_e^r + h_D + R_{>D}$ be a convergent analytic germ in three variables, where $D = er + k$, $e \geq 3$, $r \geq 2$, $k \geq 1$, the plane curve $C = V(g_e)$ is smooth, and $h_D|_C \not\equiv 0$. We prove that the local Bernstein–Sato polynomial contains the uniform factor $\prod_{j=0}^{D-1}(s + (3r+k)/(rD) + j/D)$, independently of the contact data and the higher analytic terms. Holomorphic numerators give a triangular set of exact roots containing this block. For the two-layer germs $g_e^r + h_D$, with $V(h_D)$ smooth at its intersections with C, we determine a root for every nontransverse contact order $q \geq 2$ and prove that every pole of the local topological zeta function is a root. An explicit infinite family has a contact root outside the triangular set even though the corresponding topological pole cancels. The proof combines compact residue periods with a cyclic cover of C; ramification and Riemann–Roch supply the numerators that fill the uniform factor.

1 Introduction

For surface germs whose first two homogeneous layers are $g_e^r + h_D$, with a smooth projective curve $C = V(g_e)$, we construct a uniform block of D exact Bernstein–Sato roots. This block depends only on e, r, D and persists under arbitrary higher analytic terms. Nontransverse contacts give a second source of roots. In an explicit infinite family, a contact root lies outside the entire triangular root set even though its topological pole cancels.

Let $g_e, h_D \in \mathbf{C}[x, y, z]$ be homogeneous of degrees e, D. For integers e, r, k, set

$$A = er, \qquad D = er + k, \qquad e \geq 3, \quad r \geq 2, \quad k \geq 1. \tag{1.1}$$

We assume that g_e defines a smooth plane curve $C \subset \mathbf{P}^2$ of degree e and that $h_D|_C \not\equiv 0$. The first theorem applies to every convergent germ

$$f = g_e^r + h_D + R_{>D}, \qquad \operatorname{ord}_0 R_{>D} > D. \tag{1.2}$$

Write $b_{f,0}(s)$ for its monic local Bernstein–Sato polynomial: it generates the polynomials $b(s)$ for which $b(s)f^s = P(s)f^{s+1}$ with $P(s) \in \mathcal{D}_{\mathbf{C}^3,0}[s]$. Put

$$\alpha_R = \frac{3r + k}{rD}. \tag{1.3}$$

Theorem 1.1 (Uniform factor and triangular root set)**.** *For every germ* (1.2)*,*

$$\prod_{j=0}^{D-1} \left(s + \alpha_R + \frac{j}{D} \right) \;\mid\; b_{f,0}(s). \tag{1.4}$$

More generally, for integers ℓ, j,

$$b_{f,0}\left(-\alpha_R - \frac{\ell}{r} - \frac{j}{D} \right) = 0 \quad \text{if} \quad \begin{cases} 0 \leq \ell \leq r-2, \quad 0 \leq j < D, \\ \ell/r + j/D < 1. \end{cases} \tag{1.5}$$

The factor in (1.4) depends only on e, r, k. In particular, the theorem allows arbitrary zero multiplicities of $h_D|_C$ and nonisolated germs. The triangular set (1.5) contains the uniform block and strictly enlarges it exactly when $r \geq 3$ and $r \nmid k$. It is obtained from the original holomorphic numerators $g_e^\ell p_j$, where p_j has degree j. Different pairs (ℓ, j) may give the same root; throughout the paper root sets are counted without repetition.

For the contact results, suppose that

$$f = g_e^r + h_D, \qquad H = V(h_D) \text{ is smooth at every point of } P = C \cap H. \tag{1.6}$$

Smoothness here means that the local defining equation of H has nonzero differential at each point of P. The set P is finite. Let $q_p = I_p(C, H)$ be the actual intersection multiplicity, and define

$$B = (r-1)D + k, \qquad \beta = 3(r-1) + k, \qquad \alpha_q = \frac{q\beta + k}{qB}. \tag{1.7}$$

Theorem 1.2 (Contact roots)**.** *For every contact order $q = q_p \geq 2$ in (1.6),*

$$b_{f,0}(-\alpha_q) = 0. \tag{1.8}$$

The contact formula detects candidate exponents even after they cancel from the topological zeta function. The following family places one of these roots outside the triangular set.

Theorem 1.3 (Contact roots after pole cancellation)**.** *Suppose that the integers r, e, k, q satisfy*

$$r \geq 3, \qquad k = r - 1, \qquad e \geq 3, \qquad D = r(e+1) - 1 = 4q. \tag{1.9}$$

There exist pairs (C, H) as in (1.6) with one contact of order q and all other intersections transverse. For every such pair,

$$-\frac{1}{q} \notin \operatorname{Poles} Z_{\mathrm{top},0}(f; s), \qquad b_{f,0}(-1/q) = 0. \tag{1.10}$$

The root $-1/q$ lies outside the triangular set (1.5).

An infinite subfamily has $r = 3$, $k = 2$, $e = 4m + 2$ and $q = 3m + 2$, for integers $m \geq 1$. Its first value, $(e, r, k, q) = (6, 3, 2, 5)$, has at least 37 distinct Bernstein–Sato roots; see Example 6.1.

The pole calculation gives the following application.

Corollary 1.4 (Strong topological monodromy for the two-layer family)**.** *Under (1.6), the germ f is reduced and has an isolated critical point at the origin. Every pole of $Z_{\mathrm{top},0}(f; s)$ is simple, and*

$$s_0 \in \operatorname{Poles} Z_{\mathrm{top},0}(f; s) \quad \Longrightarrow \quad b_{f,0}(s_0) = 0. \tag{1.11}$$

Period construction

The uniform factor comes from a divisor lying over C. A cyclic cover of C separates its residue into an Euler form in the fiber and a differential on a compact curve. The key step is to choose p_j so that the latter differential has a nonzero period. At a zero of $h_D|_C$ of order q_x, its regularity condition is the explicit inequality

$$\operatorname{ord}_x(p_j|_C) \geq \left\lfloor \frac{(j + 3 - e)q_x}{D} \right\rfloor.$$

The low degrees give holomorphic forms directly. For $e \geq 4$, Riemann–Roch fills the remaining degrees $j < D$. For a cubic C, a degree-zero line bundle can require a differential with simple

poles and a nonzero residue period. The corresponding original numerator is holomorphic. These constructions fill the entire block in (1.4).

At a contact point, the prescribed residue is a two-dimensional Euler form. In the exceptional case where its first parameter is $a = 1$, the contact root belongs to the curve's uniform block. For all other parameters, explicit de Rham identities give a nonzero period through a convergent positive integral or a logarithmic residue. Section 2 proves the period criterion, Section 3 constructs the uniform roots, and Section 4 proves the contact theorem. Sections 5 and 6 give the topological application and the cancellation examples.

Relation with earlier work

Partial resolutions and their zeta functions are classical tools for surface singularities; see Gusein-Zade, Luengo, and Melle-Hernández [6]. Denef and Loeser's Newton formula [4] supplies the local orbit calculation used here, with the Jacobian of the first blow-up retained. Related suspension and Lê–Yomdin formulas are developed in [1]. For the degree-gap-one family, the companion paper [7] computes the topological and monodromy invariants and reconstructs the contact orders from either. The present paper constructs exact Bernstein–Sato roots for arbitrary degree gap and a uniform factor in the broader analytic family (1.2).

Our analytic input is the residue-period argument, in the form of the all-numerator criterion of Esterov [5, Lemma 7.3]. Section 2 gives its divisor form with a holomorphic numerator.

Blanco [3, Theorem B and Proposition C] constructs geometric roots for plane curves on selected rupture divisors from holomorphic numerators with $0 \le v_i(g) < N_i$ and residue conditions, and compares the resulting roots with jumping numbers and motivic poles. For our three-variable family, ramification of the cyclic curve cover and Riemann–Roch supply original numerators and nonzero periods for every $j < D$, producing the entire uniform factor.

Examples of Bernstein–Sato roots absent from the motivic zeta pole set occur in Bath and Veys [2, Example 2.13]. Theorem 1.3 supplies an explicit infinite contact family, with a nonzero specified residue period and a root outside the triangular set (1.5).

2 Compact residue periods and exact roots

A compact period of the prescribed normal residue detects an exact Bernstein–Sato root. The criterion below keeps track of the numerator's vanishing order; Euler forms then supply the periods. All cycles have complex coefficients and finite, hence compact, support.

2.1 A divisor criterion with a holomorphic numerator

Let $\pi : Y \to (\mathbf{C}^n, 0)$ be an embedded resolution, an isomorphism outside $f^{-1}(0)$, and let E be a prime divisor with $\pi(E) = \{0\}$. On a smooth open subset E° meeting no other component of the total transform, write

$$f \circ \pi = t^N U(t, y), \qquad \pi^*\omega_0 = t^{\nu-1} J(t, y)\, dt \wedge dy, \tag{2.1}$$

where $\omega_0 = dx_1 \wedge \cdots \wedge dx_n$ and U, J are units. In the normal line bundle of E, define

$$\widehat{E}^\circ = \{(y, \tau) : \tau^N U(0, y) = 1\}. \tag{2.2}$$

This definition is intrinsic: if $t_b = c_{ba} t_a + O(t_a^2)$, then $\tau_b = c_{ba}\tau_a$ and $U_b(0, y) = c_{ba}^{-N} U_a(0, y)$.

Proposition 2.1 (Numerator period criterion)**.** *Let $p \in \mathcal{O}_{\mathbf{C}^n,0}$ and write $p \circ \pi = t^m P(t, y)$, with $0 \le m < N$. If the form*

$$\rho_p = \tau^{\nu+m} J(0, y) P(0, y)\, dy \tag{2.3}$$

has a nonzero period on a compact $(n-1)$-cycle in $\widehat{E}^\circ$, then

$$b_{f,0}\Big(-\frac{\nu+m}{N}\Big)=0. \tag{2.4}$$

Proof. Comparing the leading terms of $\pi^*(p\omega_0)$ on overlapping charts gives

$$J_b(0,y_b)P_b(0,y_b)\,dy_b = c_{ba}^{-(\nu+m)}J_a(0,y_a)P_a(0,y_a)\,dy_a.$$

Together with $\tau_b = c_{ba}\tau_a$, this proves that (2.3) is a global form on $\widehat{E}^\circ$. Fix a cycle Γ with $c=\int_\Gamma \rho_p \neq 0$. Its support is covered by finitely many charts (2.1). Choose a smooth tubular map Φ whose normal derivative along E is complex linear, and put $\epsilon = z^{1/N} > 0$. Uniformly near this support,

$$\epsilon^{-N}(f\circ\pi)(\Phi(y,\epsilon s\tau)) = s^N + O(\epsilon).$$

At $(s,\epsilon)=(1,0)$, the real two-dimensional derivative in the complex variable s is invertible. The implicit function theorem gives a unique smooth solution $s_\epsilon = 1+O(\epsilon)$ for which this expression equals one. Uniqueness makes the local solutions agree. The resulting map to $(f\circ\pi)^{-1}(z)$ satisfies

$$t=\epsilon\tau+O(\epsilon^2), \qquad y_\epsilon = y+O(\epsilon)$$

in tangential C^1, uniformly on the fixed support. It sends Γ to a closed cycle; pushing it by π gives a cycle $\gamma_z \subset f^{-1}(z)$ whose support tends uniformly to the origin.

The relative Gelfand–Leray form in these coordinates is

$$\frac{\pi^*\omega_0}{d(f\circ\pi)} = \frac{t^{\nu-N}J(t,y)}{NU(t,y)+t\partial_t U(t,y)}\,dy.$$

Its denominator is bounded away from zero on the chosen neighborhood. Substitution gives, with $\alpha_p=(\nu+m)/N$ and $\alpha_0=\nu/N$,

$$\begin{aligned} I_p(z) &:= \int_{\gamma_z}\frac{p\omega_0}{df} = \frac{1}{N}z^{\alpha_p-1}(c+O(z^{1/N})),\\ I_h(z) &:= \int_{\gamma_z}\frac{h\omega_0}{df} = O_h(z^{\alpha_0-1}) \quad (h\in\mathcal{O}_{\mathbf{C}^n,0}). \end{aligned} \tag{2.5}$$

The second estimate applies to every original holomorphic numerator h.

Take a Bernstein identity $b(s)f^s = \mathcal{P}(s)f^{s+1}$. Multiplying by $p\omega_0$ and integrating the derivatives in $\mathcal{P}(s)$ by parts gives

$$b(s)pf^s\omega_0 = f^{s+1}h_s\omega_0 + d\Xi_s, \qquad h_s=\sum_{a=0}^{L}s^a h_a, \tag{2.6}$$

Here $h_s=\mathcal{P}(s)^*p$, where the star is the formal adjoint with respect to ω_0; in particular, every h_a is holomorphic in the original space. The form Ξ_s is a finite sum of terms $f^{s+1-b}\eta_b(s)$ with holomorphic $(n-1)$-forms $\eta_b(s)$ polynomial in s. For such a form, $df\wedge\eta_b = h\omega_0$ with h holomorphic, so its fiber integral has the second bound in (2.5).

Integrate (2.6) over the chain swept out by γ_z, $0<z\le z_0$. For $\Re s$ sufficiently large the lower boundary tends to zero, and

$$b(s)M_p(s) = \sum_{a=0}^{L}s^a M_{h_a}(s+1)+B(s), \qquad M_h(s)=\int_0^{z_0}z^s I_h(z)\,dz, \tag{2.7}$$

with $B(s)$ entire. Both sides are holomorphic for $\Re s > -\alpha_p$, so the identity holds on that half-plane. The right side is moreover holomorphic for $\Re s > -\alpha_0 - 1$. Since $m < N$, it is holomorphic near $s = -\alpha_p$. The first asymptotic gives

$$\lim_{u \downarrow 0} u M_p(-\alpha_p + u) = c/N \neq 0.$$

Multiply (2.7) by u and let $s = -\alpha_p + u$ tend to $-\alpha_p$. This proves $b(-\alpha_p) = 0$, in particular for $b = b_{f,0}$. □

2.2 Euler forms with compact periods

All Euler parameters below are rational. A finite Kummer cover makes the forms single-valued. The following elementary regularization will also specify the cycles.

Lemma 2.2 (Positive Euler periods)**.** *For $m = 1, 2$, let*

$$M_m = \{z_1 \cdots z_{m+1} \neq 0, \quad z_1 + \cdots + z_{m+1} = 1\}.$$

If every λ_i is positive and nonintegral, then $\prod_i z_i^{\lambda_i - 1} dz_1 \wedge \cdots \wedge dz_m$ has a nonzero compact period on its finite Kummer cover.

Proof. Truncate the real simplex $z_i > 0$ near its faces and choose the positive branch. A face has normal monodromy $\zeta_i \neq 1$. A small normal circle has twisted boundary $\zeta_i - 1$, so dividing by this scalar supplies a tube that cancels the truncated boundary. For $m = 2$, the remaining corner boundaries are filled by two-torus chains. More explicitly, at a corner put $a = \zeta_i - 1$, $b = \zeta_j - 1$. The local boundary complex is

$$\mathbf{C} \xrightarrow{c \mapsto (-bc, ac)} \mathbf{C}^2 \xrightarrow{(x,y) \mapsto ax + by} \mathbf{C}.$$

Every cycle (x, y) in the middle is the image of y/a. Thus the face tubes and corner chains close the truncated simplex to a finite twisted cycle.

Changing the cutoff gives homologous cycles. Positivity of the parameters makes the integrals over the added tubes tend to zero with the cutoff. The cycle integral therefore equals the strictly positive convergent simplex integral. Finally, lift the chain to the finite cover and sum its lifts with the character coefficients. This produces an ordinary complex-coefficient cycle with a nonzero integral. □

Write $W = 1 - X - Y$ on M_2. The surface and curve Euler forms are

$$\Phi_{a,b,c} = X^{a-1} Y^{b-1} W^{c-1} dX \wedge dY, \qquad \varphi_{a,b} = z^{a-1}(1 - z)^{b-1} dz. \tag{2.8}$$

Lemma 2.3 (Euler nonvanishing)**.** *Each of the following forms has a nonzero compact period on its finite Kummer cover:*

(i) *$\varphi_{a,b}$, if $a \leq 0$, $b > 0$ is nonintegral, and $a + b > 0$;*

(ii) *$\Phi_{a,b,c}$, if $a, b > 0$ are nonintegral, $c \leq 0$, and $a + b + c > 0$.*

Proof. The identities

$$d(z^a(1 - z)^b) = a\varphi_{a,b} - (a + b)\varphi_{a+1,b}, \tag{2.9}$$

$$d(X^{a-1} Y^{b-1} W^c (X dY - Y dX)) = (a + b + c)\Phi_{a,b,c+1} - c\Phi_{a,b,c} \tag{2.10}$$

hold on the same cover as the original forms. If the nonpositive parameter is nonintegral, iterate until it first becomes positive. The resulting cohomology multiplier is $(\sigma)_u/(v)_u$, where

v is the initial parameter, σ is the initial sum, and $(x)_u = x(x+1)\cdots(x+u-1)$. It is nonzero. Lemma 2.2 applies to the final form.

If $v = -m$ is an integer, stop at zero. The multiplier is $(-1)^m(\sigma)_m/m!$. For the curve form at $a = 0$, a small circle around $z = 0$ has nonzero integral, since the residue is one. For the surface form at $c = 0$, the residue along $W = 0$ is, up to sign, $X^{a-1}(1-X)^{b-1}dX$. Its compact period is nonzero by Lemma 2.2. A small normal circle tube over that cycle pairs with $\Phi_{a,b,0}$ by $2\pi i$ times this period. The normal character is trivial, so the tube closes. The nonzero multipliers prove both assertions. □

Integer changes of a parameter multiply forms by single-valued functions, so all these identities take place on one cover. If a geometric cover is larger than the minimal cover of a form, transfer along the finite étale map lifts a cycle and multiplies its integral by the degree. We will use this to pass between the explicit Euler covers and the normal covers of divisors.

3 The cyclic curve cover and the uniform factor

On a finite cover, the residue over C separates into an Euler form in the fiber and a differential on a compact curve. Ramification controls the poles of the curve differential, and Riemann–Roch supplies the numerators that give its nonzero periods.

3.1 A divisor over the tangent curve

Blow up the origin. Near C, use a radial coordinate u_i, the normal coordinate $v_i = g_i$, and a curve coordinate w_i. With $h_i^0 = h_i(0, w_i)$, the pullbacks are

$$\begin{aligned} f \circ \pi_0 &= u_i^A(v_i^r + u_i^k h_i(v_i, w_i) + u_i^{k+1} Q_i(u_i, v_i, w_i)), \\ \pi_0^*\omega_0 &= u_i^2 J_i(v_i, w_i)\, du_i \wedge dv_i \wedge dw_i. \end{aligned} \tag{3.1}$$

Set $\eta_i = J_i(0, w_i)dw_i$. Along C, a projective change of scale λ gives

$$u_b = \lambda u_a, \quad v_b = \lambda^{-e} v_a, \quad h_b^0 = \lambda^{-D} h_a^0, \quad \eta_b = \lambda^{e-3}\eta_a. \tag{3.2}$$

The last relation is the adjunction formula $K_C \simeq \mathcal{O}_C(e-3)$ in these local frames.

Let $\gamma = \gcd(r, k)$, $r' = r/\gamma$, $k' = k/\gamma$. Over $C^\circ = C \setminus P$, where P is the zero set of $h_D|_C$, take a toroidal modification containing the primitive ray (r', k') in the (u, v) plane. The two Cartier divisors $u = 0$, $v = 0$ make these modifications compatible on overlaps. A regular subdivision and a resolution away from the open part under consideration give a divisor E_R with

$$N_R = \frac{rD}{\gamma}, \qquad \nu_R = \frac{3r + k}{\gamma}, \qquad \frac{\nu_R}{N_R} = \alpha_R. \tag{3.3}$$

The terms containing $h_i - h_i^0$ or Q_i have strictly larger weight than N_R.

3.2 Separating the curve and the Euler fiber

Let $p : \widetilde{C} \to C$ be the compact normalized cyclic cover defined in the line bundle $\mathcal{O}_C(1)$ by

$$H_i^D = h_i^0, \qquad H_b = \lambda^{-1} H_a. \tag{3.4}$$

The cover has degree D and may be disconnected. Above C° it is étale. On this cover put

$$U = u_i H_i, \qquad V = v_i / H_i^e.$$

The leading function is $U^A(V^r + U^k)$, and the leading volume form is

$$H_i^{e-3}U^2 dU \wedge dV \wedge p^*\eta_i. \tag{3.5}$$

Choose integers L, M with $r'M - k'L = 1$. A unimodular torus chart at E_R is

$$U = t^{r'}Z^L, \qquad V = t^{k'}Z^M. \tag{3.6}$$

Consequently,

$$\begin{aligned} f \circ \pi &= t^{N_R}Z^{LD}(1 + Z^\gamma) + \text{higher powers of } t, \\ \pi^*\omega_0 &= t^{\nu_R - 1}Z^{3L+M-1}dt \wedge dZ \wedge H_i^{e-3}p^*\eta_i + \text{higher powers of } t. \end{aligned} \tag{3.7}$$

The leading coordinates in (3.6) give the gluing of the normal cover. Extending H_i locally off C gives on overlaps $U_b/U_a = 1 + O(v_a)$ and $V_b/V_a = 1 + O(v_a)$. Since $t = U^M V^{-L}$ and $Z = U^{-k'}V^{r'}$, it follows that

$$t_b/t_a = 1 + O(t_a^{k'}), \qquad Z_b/Z_a = 1 + O(t_a^{k'}). \tag{3.8}$$

Thus Z and the first normal coordinate agree along the divisor after pullback to $\widetilde{C}^\circ = p^{-1}(C^\circ)$. For

$$\Sigma = \{(Z, \tau) : Z \neq 0,\ Z^\gamma \neq -1, \quad \tau^{N_R}Z^{LD}(1 + Z^\gamma) = 1\}, \tag{3.9}$$

we obtain a finite étale map of degree D

$$\mathcal{M} : \widetilde{C}^\circ \times \Sigma \longrightarrow \widehat{E}_R^\circ. \tag{3.10}$$

It is the base-change projection induced by the finite étale cover $\widetilde{C}^\circ \to C^\circ$.

Let p_j be homogeneous of degree j with nonzero restriction $a_j = p_j|_C$. For a nonnegative integer ℓ, use the original holomorphic numerator $g_e^\ell p_j$. Its order along E_R is

$$m_{\ell,j} = \frac{\ell D + jr}{\gamma}, \qquad \frac{m_{\ell,j}}{N_R} = \frac{\ell}{r} + \frac{j}{D}. \tag{3.11}$$

Since $g_e = u_i^e v_i = U^e V$, the curve factor of its residue is independent of ℓ. Direct substitution in (3.7) gives

$$\mathcal{M}^*\rho_{g_e^\ell p_j} = \phi_{\ell,j} \wedge \theta_j, \qquad \theta_j = H_i^{e-3-j}p^*(a_{j,i}\eta_i). \tag{3.12}$$

The transitions (3.2) and (3.4) show that θ_j is global. The fiber factor is the single-valued form on Σ

$$\phi_{\ell,j} = \tau^{\nu_R + m_{\ell,j}}Z^{(3+j+e\ell)L+(\ell+1)M-1}dZ. \tag{3.13}$$

Let $\alpha_{\ell,j} = \alpha_R + \ell/r + j/D$. Using the defining equation of Σ and $rM - kL = \gamma$, its branch expression reduces to

$$\phi_{\ell,j} = Z^{\gamma(\ell+1)/r-1}(1 + Z^\gamma)^{-\alpha_{\ell,j}}dZ \tag{3.14}$$

up to a fixed nonzero constant. Set $Y = Z^{-\gamma}$ and $z = Y/(1 + Y)$. The map $\Sigma \to M_1$, $z = 1/(1 + Z^\gamma)$, is finite étale. Under this map, (3.14) becomes a nonzero multiple of

$$\varphi_{A_j,B_\ell}, \qquad A_j = \frac{3 + j - e}{D}, \quad B_\ell = \frac{\ell + 1}{r}, \quad A_j + B_\ell = \alpha_{\ell,j} > 0. \tag{3.15}$$

The single-valued expression (3.13) shows that the Euler character becomes trivial on Σ. Thus each connected component maps to the minimal character cover of φ_{A_j,B_ℓ}. Transfer to one such component lifts any nonzero period to Σ.

3.3 Ramification supplies the numerators

The fiber period is nonzero for every pair in (1.5): when $A_j \le 0$, use Lemma 2.3(i); when $A_j > 0$, both A_j and B_ℓ lie in $(0,1)$, so use Lemma 2.2. We now construct a nonzero period of θ_j for every $0 \le j < D$.

For $j \le e-3$, take any nonzero homogeneous p_j. Its restriction to C is nonzero because $j < e$, and θ_j is a nonzero holomorphic differential on $\widetilde{C}$. It therefore has a nonzero period, which can be represented away from the finitely many deleted points.

For $j \ge e-2$, put $v = j+3-e > 0$ and write

$$\operatorname{div}_C(h_D|_C) = \sum_{x \in P} q_x x, \qquad \sum_x q_x = eD.$$

Set $d_x = \gcd(D, q_x)$. On a normalized branch over x, a local parameter z satisfies $w = z^{D/d_x} \cdot \text{unit}$ and $H = z^{q_x/d_x} \cdot \text{unit}$. If a_j vanishes to order n at x, then

$$\operatorname{ord}_z \theta_j = \frac{D(n+1) - vq_x - d_x}{d_x}. \tag{3.16}$$

The ramification contribution $D/d_x - 1$ from $p^*\eta_i$ is included here. Because the remainder of vq_x modulo D is a multiple of d_x, this order is nonnegative exactly when

$$n \ge \left\lfloor \frac{vq_x}{D} \right\rfloor. \tag{3.17}$$

Define

$$\Gamma_v = \sum_x \left\lfloor \frac{vq_x}{D} \right\rfloor x, \qquad L_j = \mathcal{O}_C(j)(-\Gamma_v). \tag{3.18}$$

With $g_C = (e-1)(e-2)/2$,

$$\deg L_j \ge ej - ev = e(e-3) = 2g_C - 2. \tag{3.19}$$

If $e \ge 4$, Riemann–Roch gives $h^0(C, L_j) \ge g_C - 1 > 0$. A nonzero section a_j produces a nonzero holomorphic θ_j on the compact normalized cover, by (3.16). It lifts to a homogeneous polynomial p_j: the restriction sequence

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}^2}(j-e) \longrightarrow \mathcal{O}_{\mathbf{P}^2}(j) \longrightarrow \mathcal{O}_C(j) \longrightarrow 0 \tag{3.20}$$

is surjective on sections, since $H^1(\mathbf{P}^2, \mathcal{O}_{\mathbf{P}^2}(n)) = 0$ for every integer n.

For $e = 3$, the curve is elliptic and $v = j$. A positive-degree L_j has a section by Riemann–Roch; the same construction applies whenever $H^0(C, L_j) \ne 0$. It remains to treat $\deg L_j = 0$ and $H^0(C, L_j) = 0$. Equality in (3.19) then forces every jq_x/D to be an integer, say n_x. Since $j > 0$, some $n_{x_0} \ge 1$. The degree-one bundle $L_j(x_0)$ has a one-dimensional section space. A nonzero section cannot belong to $H^0(C, L_j) = 0$. Thus, as a section of $\mathcal{O}_C(j)$, it vanishes to order exactly $n_{x_0} - 1$ at x_0 and at least n_x at the other points. Here $\Gamma_j - x_0$ is effective, so the section remains holomorphic on the original curve and lifts by (3.20). Formula (3.16) gives simple poles over x_0 and holomorphy elsewhere. At each such pole its expansion is $(c_z/z + O(1))dz$ with $c_z \ne 0$, so a small circle has the nonzero period $2\pi i c_z$.

In all cases we have an original polynomial p_j and a compact cycle $\gamma_j \subset \widetilde{C}^\circ$ with $\int_{\gamma_j} \theta_j \ne 0$.

3.4 From product periods to roots

For a pair (ℓ, j) in (1.5), choose the numerator above and a nonzero fiber cycle $\delta_{\ell,j}$ from (3.15). Transfer to Σ if necessary. By (3.12), the pushforward

$$\mathcal{M}_*(\gamma_j \times \delta_{\ell,j})$$

has nonzero pairing with the specified residue, up to the fixed orientation sign. Equation (3.11) gives $m_{\ell,j} < N_R$, so Proposition 2.1 yields $b_{f,0}(-\alpha_{\ell,j}) = 0$. Taking $\ell = 0$ gives D distinct roots and hence the divisibility (1.4). This proves Theorem 1.1.

For $r = 2$, the triangular set has only $\ell = 0$. If $r \mid k$, then D/r is an integer and the reindexing $j' = j + \ell D/r$ puts every triangular root in the uniform block: the window condition gives $0 \le j' < D$. For $r \ge 3$ and $r \nmid k$, the allowed pair $(\ell, j) = (1, 0)$ gives a root outside that block, since $D/r \notin \mathbf{Z}$.

4 Contact roots

Assume (1.6) and fix $p \in P$ with contact order $q \ge 2$. We construct the contact period explicitly.

4.1 The local model and its normal cover

In (3.1), smoothness of H gives $h(v, w) = a(v, w)(v + w^q)$ after a change of w, where a is a unit. Replacing the radial coordinate by $ua^{1/k}$ puts the function and the original volume form, up to analytic units, in the form

$$F = u^A(v^r + u^k v + u^k w^q), \qquad \omega = u^2 du \wedge dv \wedge dw. \tag{4.1}$$

Let

$$T' = (q(r-1), qk, k), \quad t_q = \gcd(k, q(r-1)), \quad T = T'/t_q. \tag{4.2}$$

The compact Newton face is nondegenerate. A regular toric subdivision containing T gives a divisor E_T with

$$N_T = qB/t_q, \qquad \nu_T = (q\beta + k)/t_q, \qquad \nu_T/N_T = \alpha_q. \tag{4.3}$$

All three weights of T are positive. Thus the analytic units in the model restrict to nonzero constants on its divisor torus. Denote the function and volume constants by $c_f, c_\omega \ne 0$, respectively.

To compute that residue, use the finite monomial map

$$u = t^{q(r-1)}\xi^{-1}\eta^{-(r-1)}, \qquad v = t^{qk}\eta^{-k}, \qquad w = t^k. \tag{4.4}$$

Its exponent matrix has determinant of absolute value k^2, and direct substitution gives

$$\begin{aligned} F &= t^{qB}\xi^{-D}\eta^{-B}(1 + \xi^k + \eta^k), \\ \omega &= \pm k^2 t^{q\beta+k-1}\xi^{-4}\eta^{-\beta-1} dt \wedge d\xi \wedge d\eta. \end{aligned} \tag{4.5}$$

The map factors through a chart of the primitive ray T. On the divisor torus it is finite étale, with normal ramification index t_q. A normal coordinate t_0 at E_T satisfies $t_0 = t^{t_q} M(\xi, \eta)$ for an invertible monomial M. Define the auxiliary normal cover by

$$\widehat{M}_T : \quad \vartheta^{qB}\xi^{-D}\eta^{-B}(1 + \xi^k + \eta^k) = 1, \qquad \xi, \eta, \vartheta \ne 0. \tag{4.6}$$

Choose c_1 with $c_1^{N_T} = c_f^{-1}$. There is a finite étale map $\kappa : \widehat{M}_T \to \widehat{E}_T^\circ$ given by $\tau_0 = c_1 M \vartheta^{t_q}$. The specified single-valued form on the auxiliary cover is

$$\rho' = \pm k^2 \vartheta^{q\beta+k}\xi^{-4}\eta^{-\beta-1} d\xi \wedge d\eta. \tag{4.7}$$

The auxiliary orders are $t_q N_T$ and $t_q \nu_T$. Using $dt_0/t_0 = t_q\, dt/t + d\log M$, the residue computed in (4.7) satisfies

$$\rho' = t_q c_\omega^{-1} c_1^{-\nu_T} \kappa^* \rho_{E_T}, \tag{4.8}$$

up to the common orientation sign.

Set $x = \xi^k$, $y = \eta^k$. The factor k^2 cancels the Jacobian of this change, and the residue becomes

$$x^{a-1}y^{b-1}(1+x+y)^{-\alpha_q}dx \wedge dy, \qquad a = \frac{D\alpha_q - 3}{k}, \quad b = \frac{B\alpha_q - \beta}{k}.$$

The projective change $X = x/(1+x+y)$, $Y = y/(1+x+y)$, $W = 1/(1+x+y)$ gives $\Phi_{a,b,c}$ with

$$\begin{aligned} a &= \frac{D - 3 + D/q}{B} > 0, \qquad b = \frac{1}{q}, \\ c &= -\frac{r[q(e-3)+D]}{qB} < 0, \qquad a+b+c = \alpha_q > 0. \end{aligned} \tag{4.9}$$

These substitutions define a finite étale map $\widehat{M}_T \to M_2$. Equation (4.7) shows that it lifts to the minimal Euler character cover. Transfer followed by κ_* therefore preserves nonvanishing of periods, by (4.8).

4.2 Proof of the contact root theorem

If $r \geq 3$, then $0 < a < 1$. If $r = 2$, then $0 < a < 3/2$; its only possible integral value is $a = 1$. This value occurs exactly when

$$r = 2, \qquad D = q(k+3), \qquad (a,b,c) = (1, 1/q, -1), \qquad \alpha_q = 1/q. \tag{4.10}$$

The identity $2e = (q-1)k + 3q$ forces k, q to be even. For $j = k/2$,

$$e - 3 - j = \frac{(q-2)(k+3)}{2} \geq 0, \qquad \alpha_R + \frac{j}{D} = \frac{k+3}{D} = \frac{1}{q}.$$

Theorem 1.1 gives $b_{f,0}(-\alpha_q) = 0$. The associated torus form has the explicit primitive

$$\Phi_{1,b,c} = d\left(\frac{Y^b W^{c-1}}{b}\, dW\right), \qquad X = 1 - Y - W. \tag{4.11}$$

For all other parameters, $a, b > 0$ are nonintegral and $c < 0$. Lemma 2.3(ii) supplies a nonzero compact Euler period. Transfer followed by κ_* gives a nonzero period of ρ_{E_T}, and Proposition 2.1 with numerator 1 gives $b_{f,0}(-\alpha_q) = 0$. This completes the proof of Theorem 1.2.

5 Topological poles and strong monodromy

For the two-layer family (1.6), the topological zeta function is determined by e, r, k and the contact orders. We compute it and identify its poles among the roots constructed above.

These hypotheses make f reduced with an isolated critical point at the origin. Indeed, if $f = df = 0$ at a nonzero point, the homogeneous Euler identity gives

$$Ag_e^r + Dh_D = 0, \qquad g_e^r + h_D = 0.$$

Thus $g_e = h_D = 0$, where $df = dh_D \neq 0$ by smoothness of H at P, a contradiction. A positive-dimensional critical branch through the origin would have f constant and hence zero, so the critical point is isolated. A repeated factor would produce a positive-dimensional critical set, proving reducedness.

Let $(E_i)_{i \in J}$ be the components of the function's total transform on a resolution π, including the strict transforms. Their numerical data are

$$N_i = \operatorname{ord}_{E_i}(f \circ \pi), \qquad \nu_i - 1 = \operatorname{ord}_{E_i}(\pi^* \omega_0).$$

For $\varnothing \neq I \subseteq J$, set

$$E_I^\circ = \left(\bigcap_{i\in I} E_i\right) \setminus \bigcup_{j\notin I} E_j.$$

Our convention is

$$Z_{\mathrm{top},0}(f;s) = \sum_{\varnothing\neq I\subseteq J} \chi(E_I^\circ \cap \pi^{-1}(0)) \prod_{i\in I} \frac{1}{N_i s + \nu_i}. \tag{5.1}$$

Here $\omega_0 = dx \wedge dy \wedge dz$.

5.1 The complete zeta function

Write $\chi_C = e(3-e)$, $b = |P|$, and set

$$\begin{gathered} a = As+3, \qquad d = s+1, \qquad \mathcal{R} = rDs + 3r + k, \\ K_q = qBs + q\beta + k, \qquad \gamma = \gcd(r,k), \quad h = \gcd(k, r-1), \\ j_q = \gcd(k,r,q), \qquad t_q = \gcd(k, q(r-1)), \\ G_q = \mathcal{R}(t_q^2 - qh^2) - k(j_q^2 + q(r-1) - r). \end{gathered} \tag{5.2}$$

Proposition 5.1 (Weighted zeta formula)**.** *For* (1.6)*,*

$$\begin{aligned} Z_{\mathrm{top},0}(f;s) = {} & \frac{3-\chi_C}{a} + \frac{k\chi_C}{a\mathcal{R}} \\ & + \frac{r\chi_C + (\chi_C - b)(r-\gamma^2)s}{d\mathcal{R}} + \frac{s}{d\mathcal{R}} \sum_{p\in P} \frac{G_{q_p}}{K_{q_p}}. \end{aligned} \tag{5.3}$$

Consequently, every pole is simple and belongs to

$$\left\{-1, -\frac{3}{A}, -\alpha_R\right\} \cup \{-\alpha_q : q = q_p \geq 2\}. \tag{5.4}$$

Proof. After the first blow-up, the three types of strata have local models

$$u^A \quad \text{on } \mathbf{P}^2 \setminus C, \qquad G_k = u^A(v^r + u^k) \quad \text{on } C \setminus P,$$

and $F_{q,k} = u^A(v^r + u^k v + u^k w^q)$ at a point of contact order q. At a transverse intersection one obtains the same model with $q = 1$ by taking $w = h - v$. All use the volume weight u^2. The local Newton calculation in Appendix A gives

$$\begin{gathered} Z_{G_k} = \frac{k/a + r - \gamma^2 s/d}{\mathcal{R}}, \qquad Z_T = \frac{k/a + r/d}{\mathcal{R}}, \\ Z_{F_{q,k}} = Z_T + \frac{sG_q}{d\mathcal{R}K_q}. \end{gathered} \tag{5.5}$$

In particular $G_1 = 0$. Euler integration over the strata of the first exceptional plane gives

$$Z_{\mathrm{top},0}(f;s) = \frac{3-\chi_C}{a} + (\chi_C - b) Z_{G_k} + \sum_{p\in P} Z_{F_{q_p,k}}. \tag{5.6}$$

This is additivity of the finite sum (5.1), followed by Euler integration along the map to the first exceptional divisor. Substituting (5.5) proves (5.3).

The denominator gives (5.4). For the relative positions of its values, observe that

$$\alpha_R - \frac{3}{A} = \frac{k(e-3)}{erD}, \qquad \alpha_q - \alpha_R = \frac{k[q(e-3)+D]}{qDB} > 0. \tag{5.7}$$

The numbers $\alpha_q = \beta/B + k/(qB)$ are distinct for distinct $q \geq 2$, and $\alpha_R < \alpha_q < 1$. For $e \geq 4$ all entries of (5.4) are distinct. For $e = 3$, we have $a = 3(rs+1)$ and $\mathcal{R} = D(rs+1)$; the only term that could have a double pole is $k\chi_C/(a\mathcal{R})$, which vanishes since $\chi_C = 0$. Thus every pole is simple. □

5.2 The first-divisor root

The first exceptional divisor gives the root $-3/A$ throughout the analytic family (1.2).

Proposition 5.2. *Every germ* (1.2) *satisfies* $b_{f,0}(-3/A) = 0$.

Proof. The first exceptional divisor has open part $\mathbf{P}^2 \setminus C$ and data $(A, 3)$. One component of its normal cover is $\tau^e g_i = 1$. Setting $T_i = \tau^{-1}$ identifies it with $U = S \setminus B_0$, where

$$S = \{T^e = g_e(X, Y, Z)\} \subset \mathbf{P}^3, \qquad B_0 = \{T = 0\} \simeq C.$$

The projection $p_0 : S \to \mathbf{P}^2$ is finite, and S is smooth. With Ω_i the projective chart area form, its specified residue is $\sigma = p_0^*\Omega_i/T_i^3$. The numerator and denominator both change by λ^{-3} on overlaps, so σ is global. Near B_0, write $\Omega_i = J(v, w)dv \wedge dw$ and $v = T_i^e$. Then

$$\sigma = eT_i^{e-4} J(T_i^e, w)dT_i \wedge dw. \tag{5.8}$$

For $e \geq 4$, this is a nonzero holomorphic two-form on the compact surface S, with $\int_S \sigma \wedge \overline{\sigma} > 0$. If its class vanished on U, the localization sequence

$$H^0(B_0, \mathbf{C}) \longrightarrow H^2(S, \mathbf{C}) \longrightarrow H^2(U, \mathbf{C})$$

would express $[\sigma]$ as a linear combination of the Poincaré duals of the components of B_0. Pairing with $\overline{\sigma}$ gives zero, since its restriction to each curve is zero, a contradiction. Thus σ has a nonzero compact period in U. For $e = 3$, (5.8) is logarithmic with residue 3η, where η is a nonzero holomorphic differential on the elliptic curve C. A normal circle tube over a nonzero period of η gives the required period. Proposition 2.1 with numerator 1 finishes the proof. □

5.3 From poles to Bernstein–Sato roots

By Proposition 5.1, each actual pole is listed in (5.4). Proposition 5.2 handles $-3/A$, Theorem 1.1 with $j = 0$ handles $-\alpha_R$, and Theorem 1.2 handles the contact values. Finally, $b_{f,0}(-1) = 0$: substituting $s = -1$ in a Bernstein identity makes its right side $P(-1)1$ holomorphic, whereas $b(-1)/f$ cannot be holomorphic unless $b(-1) = 0$. This proves Corollary 1.4.

6 Roots beyond the pole set

In the family below, a contact root has a nonzero period even though its topological candidate cancels. We then exhibit an actual pole detected by the curve period.

6.1 An infinite family with a cancelled contact pole

Assume (1.9). Fix a smooth degree-e curve C and a point $x_0 \in C$. There is a section of $L = \mathcal{O}_C(D)$ with a zero of order exactly q at x_0 and only simple zeros elsewhere. Indeed,

$$\deg L - q - 2 = eD - q - 2 > 2g_C - 2.$$

Riemann–Roch makes one additional vanishing condition at x_0 a proper hyperplane in $H^0(C, L(-qx_0))$. For each $x \neq x_0$, a double zero at x imposes two independent conditions. The incidence space of these sections as x varies on a curve has codimension at least one. A general section has the stated divisor. Lift it to h_D using (3.20). Adding $g_e a_{D-e}$ adjusts the normal derivative at x_0 freely, so H can be chosen smooth there. At the other zeros, simplicity along C already implies smoothness of H. This proves geometric existence.

For these parameters,

$$B = k(4q+1), \qquad \beta = 4k, \qquad \mathcal{R} = 4rqs + 4r - 1, \qquad K_q = k(4q+1)(qs+1).$$

The lattice indices are $h = t_q = k$ and $j_q = 1$. Hence

$$G_q = -k^2(q-1)(\mathcal{R}+1), \qquad \frac{G_q}{K_q} = -\frac{4rk(q-1)}{4q+1}. \tag{6.1}$$

The contact term in (5.3) has lost its K_q denominator. At $s = -1/q$ the other denominators satisfy

$$a = \frac{r-1-q}{q} \neq 0, \qquad \mathcal{R} = -1, \qquad d = 1 - 1/q \neq 0;$$

here $q \geq r$. All remaining contacts are transverse and have $G_1 = 0$. Thus the complete zeta function is holomorphic at $-1/q$.

The primitive contact ray is $T = (q, q, 1)$, with

$$(N_T, \nu_T) = (q(4q+1), 4q+1).$$

Its prescribed Euler parameters are

$$(a, b, c) = \left(\frac{1}{r-1}, \frac{1}{q}, -\frac{1}{r-1}\right).$$

Equation (2.10) gives on the same cover

$$[\Phi_{1/(r-1),1/q,-1/(r-1)}] = -\frac{r-1}{q}[\Phi_{1/(r-1),1/q,1-1/(r-1)}]. \tag{6.2}$$

All three parameters on the right are positive and nonintegral, so its compact period is nonzero. The comparison (4.8) gives a nonzero period of the contact residue, and Proposition 2.1 gives $b_{f,0}(-1/q) = 0$.

Finally,

$$\frac{1}{q} - \alpha_R = \frac{1}{rD}.$$

Membership in (1.5) would require $\ell D + rj = 1$ for nonnegative integers ℓ, j, which is impossible. This proves Theorem 1.3.

Example 6.1 (At least thirty-seven distinct roots)**.** Take $(e, r, k, q) = (6, 3, 2, 5)$ and any pair constructed above. The triangular roots are

$$\left\{-\frac{11+3j}{60} : 0 \leq j \leq 19\right\} \quad \cup \quad \left\{-\frac{31+3j}{60} : 0 \leq j \leq 13\right\}.$$

The two numerator progressions differ modulo three, so there are 34 values. The roots $-1/6$, $-1/5$, and -1 lie outside this set. Thus $\deg b_{f,0} \geq 37$, although $-1/5$ is absent from the topological pole set.

6.2 An actual pole with an exact contact torus form

The following integral-coefficient example realizes the resonance (4.10). Put

$$\begin{gathered} g = x^4 + y^4 + z^4, \quad p_8 = x^8 + y^8 + z^8, \quad p_6 = x^6 + y^6 + z^6, \\ f = g^2 + z^2 p_8 + g p_6. \end{gathered} \tag{6.3}$$

Here $(e, r, k, D) = (4, 2, 2, 10)$. The four points of $C \cap \{z = 0\}$ have contact order two, since $p_8 \neq 0$ there. At these points $p_6 = y^4(y^2 - x^2) \neq 0$, so $H = V(z^2 p_8 + g p_6)$ is smooth. The remaining intersections are the 32 transverse points of $C \cap V(p_8)$. To check transversality, all coordinates at these points are nonzero, and the étale map $[x : y : z] \mapsto [x^4 : y^4 : z^4]$ reduces the intersection to the transverse line and conic $U + V + W = 0$, $U^2 + V^2 + W^2 = 0$. Thus $n_2 = 4$, $n_1 = 32$, $b = 36$, and $\chi_C = -4$.

With $a = 8s + 3$, $\mathcal{R} = 20s + 8$, $d = s + 1$, and $K_2 = 24s + 12$, formula (5.3) becomes

$$Z_{\mathrm{top},0}(f; s) = \frac{7}{a} - \frac{8}{a\mathcal{R}} + \frac{80s - 8}{d\mathcal{R}} + \frac{4s(40s + 8)}{d\mathcal{R}K_2}. \tag{6.4}$$

Its pole at $-1/2$ has residue -1. The corresponding contact torus form has parameters $(1, 1/2, -1)$ and is exact by (4.11). On E_R, the numerator $p = x$ has order $r/\gamma = 1$, with $\gamma = \gcd(r, k) = 2$ and $(N_R, \nu_R) = (10, 4)$. Its fiber form has parameters $(0, 1/2)$, and its curve form is the nonzero holomorphic differential $p^*(x_i \eta_i)$. These two periods detect the exact root $-(4 + 1)/10 = -1/2$. Theorem 1.1 gives in total the ten roots

$$-\frac{2}{5}, -\frac{1}{2}, -\frac{3}{5}, -\frac{7}{10}, -\frac{4}{5}, -\frac{9}{10}, -1, -\frac{11}{10}, -\frac{6}{5}, -\frac{13}{10};$$

the first-divisor root $-3/8$ is an additional one.

6.3 A degree bound from distinct roots

The uniform factor, the first-divisor root and the distinct contact roots give a lower bound for the degree.

Corollary 6.2. *For every germ (1.2),*

$$\deg b_{f,0} \geq D + \mathbf{1}_{e \geq 4} + \mathbf{1}_{r \nmid k}. \tag{6.5}$$

Under (1.6), let Q be the set of distinct contact orders $q \geq 2$ and put

$$\delta_q = D(\alpha_q - \alpha_R) = \frac{k[q(e - 3) + D]}{qB}.$$

Then the right side of (6.5) can be increased by $\#\{q \in Q : \delta_q \notin \mathbf{Z}\}$.

Proof. For $e \geq 4$, the first-divisor root lies above the entire uniform block by (5.7). The block contains -1 exactly when $D(1 - \alpha_R) = D - 3 - k/r$ is an integer, equivalently $r \mid k$. Finally, $\alpha_R < \alpha_q < 1$ implies that the contact root is in the block exactly when δ_q is an integer. Distinct contact orders have distinct roots. □

For $k = 1$, one has $0 < \delta_q < 1$ for every actual $q \geq 2$. Thus every distinct nontransverse contact order adds a root outside the uniform factor.

A The weighted Newton calculation

This appendix proves (5.5), keeping the volume factor u^2 from the first blow-up. The notation is that of (5.2).

A.1 Faces, cones, and lattice indices

The Newton vertices of $F_{q,k} = u^A(v^r + u^k v + u^k w^q)$ are

$$V_0 = (A, r, 0), \qquad V_1 = (D, 1, 0), \qquad V_2 = (D, 0, q).$$

Besides the coordinate normals, use the integer rays

$$\mathsf{R} = (r, k, 0), \quad \mathsf{Q} = (0, q, 1), \quad \mathsf{T} = (q(r-1), qk, k).$$

For a lattice vector v, the numerical functions are

$$N(v) = \min_i \langle v, V_i \rangle, \qquad \nu(v) = 3v_1 + v_2 + v_3.$$

Thus their linear forms $sN + \nu$ are $\mathcal{R}$, $P_q = qs + q + 1$, and K_q. Ray rescaling multiplies the cone multiplicity and its denominator by the same factor. Thus the displayed rays compute the cone quotients directly; divisor data use primitive generators, as in (4.3).

All compact faces are nondegenerate on their tori. The edges are binomials with normalized lengths $h, j_q, 1$, respectively. The triangle has normalized area t_q. Its curve maps by an isogeny of degree t_q to $1 + x + y = 0$, so its Euler characteristic is $-t_q$. A regular Newton subdivision therefore gives an actual resolution over the local model.

The weighted Denef–Loeser formula [4] is obtained from the toric orbits as follows. For a simplicial cone Δ with primitive generators v_i, write $J_\Delta = \mathrm{mult}(\Delta)/\prod_i (sN(v_i) + \nu(v_i))$. This expression is additive under subdivisions: on the saturated span it is the cone integral of $\exp(-sN - \nu)$. The coordinate denominators for the standard basis vectors e_1, e_2, e_3 are $a, 1, 1$. The coordinate boundaries for $v = 0$ and $w = 0$ have $(N, \nu) = (0, 1)$; they refine the orbit stratification and contribute denominator factor 1. Write J_τ for the cone sum attached to a compact face τ. The orbit calculation gives

$$Z_{F_{q,k}} = \sum_{\dim \tau = 0} J_\tau + \frac{s}{d} \sum_{\dim \tau \geq 1} (-1)^{\dim \tau} \mathrm{Vol}_{\mathbf{Z}}(\tau) J_\tau. \tag{A.1}$$

Here normalized volume is lattice length for an edge and twice lattice area for a triangle. Indeed, the face hypersurface V_τ in its effective torus has $\chi(V_\tau) = (-1)^{\dim \tau - 1} \mathrm{Vol}_{\mathbf{Z}}(\tau)$. The torus has Euler characteristic zero. Its complement and its intersection with the strict transform therefore give the combined coefficient

$$-\chi(V_\tau) + \frac{\chi(V_\tau)}{d} = (-1)^{\dim \tau} \mathrm{Vol}_{\mathbf{Z}}(\tau) \frac{s}{d}.$$

A vertex contributes one. Orbits with an extra torus factor have Euler characteristic zero. This proves (A.1) with the original volume weight.

The vertex cones admit the decompositions

$$(e_1, e_3, \mathsf{T}), (e_1, \mathsf{T}, \mathsf{R}); \qquad (e_3, \mathsf{Q}, \mathsf{T}); \qquad (e_2, \mathsf{R}, \mathsf{T}), (e_2, \mathsf{T}, \mathsf{Q}).$$

Their determinants have absolute values $qk, k^2, q^2(r-1), rk, q(r-1)$ in that order. For the edge cones, the gcds of the cross-product coordinates are $qh, kj_q, q(r-1)$; the ray T has lattice index t_q. Multiplying by the face volumes gives the following table. Only the edge and triangle rows are to be multiplied by s/d.

Face	Contribution before the edge/triangle factor s/d
V_0	$\dfrac{qk}{aK_q}+\dfrac{k^2}{a\mathcal{R}K_q}$
V_1	$\dfrac{q^2(r-1)}{P_qK_q}$
V_2	$\dfrac{rk}{\mathcal{R}K_q}+\dfrac{q(r-1)}{P_qK_q}$
V_0V_1	$-\dfrac{qh^2}{K_q}$
V_0V_2	$-\dfrac{kj_q^2}{\mathcal{R}K_q}$
V_1V_2	$-\dfrac{q(r-1)}{P_qK_q}$
$V_0V_1V_2$	$\dfrac{t_q^2}{K_q}$

A.2 Local simplification and the noncompact candidate

The P_q terms combine as

$$\frac{q(r-1)}{P_qK_q}\left(q+1-\frac{s}{d}\right)=\frac{q(r-1)}{dK_q}.$$

Consequently,

$$\begin{aligned}Z_{F_{q,k}}=\frac{1}{K_q}\Big[&\frac{qk}{a}+\frac{k^2}{a\mathcal{R}}+\frac{rk}{\mathcal{R}}+\frac{q(r-1)}{d}\\&+\frac{s}{d}\left(t_q^2-qh^2-\frac{kj_q^2}{\mathcal{R}}\right)\Big].\end{aligned}\tag{A.2}$$

The two-dimensional version of the same calculation for $G_k=u^A(v^r+u^k)$ has vertex terms $k/(a\mathcal{R})$ and $r/\mathcal{R}$, and edge term $-\gamma^2 s/(d\mathcal{R})$. Hence

$$Z_{G_k}=\frac{k/a+r-\gamma^2 s/d}{\mathcal{R}}.\tag{A.3}$$

For $q=1$ one has $j_1=1$, $t_1=h$, and (A.2) reduces to $Z_T=(k/a+r/d)/\mathcal{R}$. For arbitrary q, use

$$K_q=q(\mathcal{R}-a)+k,\qquad \mathcal{R}-ra=k(rs+1)$$

in (A.2). Common denominators give

$$Z_{F_{q,k}}=Z_T+\frac{s\{\mathcal{R}(t_q^2-qh^2)-k(j_q^2+q(r-1)-r)\}}{d\mathcal{R}K_q}.\tag{A.4}$$

Equations (A.3) and (A.4) are exactly (5.5). For $k=1$, all displayed gcds equal one and $G_q=-(q-1)r(Ds+4)$, recovering the contact replacement formula of [7].

Acknowledgments

This mathematical paper was prepared with the assistance of artificial intelligence. I first acknowledge the textual data produced by humanity as a whole, which provides the linguistic

and cultural record from which mathematical expression can be learned. I then thank the accumulated scholarly achievements of mathematicians throughout history. I also thank the researchers whose work has advanced and refined neural networks through successive generations. Finally, I thank the OpenAI team for the training and development of the models used in the preparation of this paper.

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